

Project Initialization and Planning Phase

Date	31 May 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Reference: <https://miro.com/templates/customer-problem-statement/>

I am	Describe the customer with 3-4 key characteristics - who are they? Smallholder precision farmers and agricultural policymakers managing fluctuating production zones.
I'm trying to	List the thing they are trying to achieve here: Identify and cultivate the most optimal crop options per plot location to secure maximized yield.
but	Describe the problems or barriers that get in the way here: Traditional guesswork remains highly unscientific, making it hard to align parameter values accurately.
because	Describe the reason the problems or barriers exist: Localized N-P-K nutrient elements and unpredictable weather fluctuations deviate significantly across planting cycles.
which makes me feel	Describe the emotions the result from experiencing the problems or barriers: Anxious, vulnerable to structural production risks, and financially restricted.

Example:

I am a farmer	I'm trying to maximize yield	But nutrients vary	Because no data engine	Which makes me feel Frustrated
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1 (Farmer)	A small-scale traditional farmer.	Select and cultivate the perfect crop matching my field characteristics for better farming efficiency.	I rely on historic guesswork and standard localized rules of thumb.	Soil variables (N, P, K, pH) and macro climate parameters alter unpredictably each year.	Anxious, helpless, and vulnerable to asset loss.
PS-2 (Researcher)	An agricultural policymaker or researcher.	Analyze crop-environment interactions to formulate data-driven optimization strategies.	Existing regional diagnostic profiles are highly unstructured and siloed.	No centralized automated validation system tracks multi-variable nutrient trends concurrently.	Restricted, reactive, and strategically inefficient.